

APSPDCL · COMMERCIAL ROOFTOP

700 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

Net investment ₹2.53 Cr after input tax credit, generating 10,73,100 units in year one against an energy charge of ₹7.65 per unit.

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹5.12 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	700 kWp
Year 1 generation	10,73,100 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹76.83 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.66 Cr
GST at 5% — recovered as ITC	₹12.67 L
Net capitalised cost	₹2.53 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹63.76 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,53,33,333	1,01,33,333	1,52,00,000	25,50,357
2	1,52,00,000	60,80,000	91,20,000	15,30,214
3	91,20,000	36,48,000	54,72,000	9,18,129
4	54,72,000	21,88,800	32,83,200	5,50,877
5	32,83,200	13,13,280	19,69,920	3,30,526
6	19,69,920	7,87,968	11,81,952	1,98,316
7	11,81,952	4,72,781	7,09,171	1,18,990
8	7,09,171	2,83,668	4,25,503	71,394
9	4,25,503	1,70,201	2,55,302	42,836
10	2,55,302	1,02,121	1,53,181	25,702
11	1,53,181	61,272	91,909	15,421
12	91,908	36,763	55,145	9,253
Tax shield, years 1-12				63,62,014

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

700 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	10,73,100	76,83,396	—	1,01,33,333	(6,16,600)	82,99,996	82,99,996
2	10,67,735	79,50,778	4,20,000	60,80,000	3,65,132	71,65,646	1,54,65,642
3	10,62,396	82,27,465	4,20,000	36,48,000	10,46,854	67,60,611	2,22,26,253
4	10,57,084	85,13,781	4,20,000	21,88,800	14,86,166	66,07,615	2,88,33,868
5	10,51,798	88,10,061	4,20,000	13,13,280	17,81,084	66,08,976	3,54,42,844
6	10,46,539	91,16,651	4,20,000	7,87,968	19,90,457	67,06,193	4,21,49,037
7	10,41,307	94,33,910	4,20,000	4,72,781	21,49,631	68,64,279	4,90,13,316
8	10,36,100	97,62,210	4,20,000	2,83,668	22,79,854	70,62,356	5,60,75,672
9	10,30,920	1,01,01,935	4,20,000	1,70,201	23,93,913	72,88,022	6,33,63,694
10	10,25,765	1,04,53,483	4,20,000	1,02,121	24,99,525	75,33,957	7,08,97,651
11	10,20,636	1,08,17,264	4,20,000	61,272	26,01,362	77,95,901	7,86,93,552
12	10,15,533	1,11,93,704	4,20,000	36,763	27,02,273	80,71,431	8,67,64,983
13	10,10,455	1,15,83,245	4,20,000	22,058	28,04,014	83,59,231	9,51,24,214
14	10,05,403	1,19,86,342	4,20,000	13,235	29,07,686	86,58,656	10,37,82,870
15	10,00,376	1,24,03,467	4,20,000	7,941	30,14,000	89,69,467	11,27,52,337
16	9,95,374	1,28,35,108	4,20,000	4,765	31,23,435	92,91,673	12,20,44,010
17	9,90,397	1,32,81,769	4,20,000	2,859	32,36,331	96,25,439	13,16,69,449
18	9,85,445	1,37,43,975	4,20,000	1,715	33,52,946	99,71,029	14,16,40,478
19	9,80,518	1,42,22,265	4,20,000	1,029	34,73,495	1,03,28,770	15,19,69,248
20	9,75,616	1,47,17,200	4,20,000	617	35,98,164	1,06,99,036	16,26,68,284
21	9,70,738	1,52,29,359	4,20,000	370	37,27,126	1,10,82,233	17,37,50,517
22	9,65,884	1,57,59,340	4,20,000	222	38,60,549	1,14,78,791	18,52,29,308
23	9,61,054	1,63,07,766	4,20,000	133	39,98,599	1,18,89,166	19,71,18,474
24	9,56,249	1,68,75,276	4,20,000	80	41,41,444	1,23,13,832	20,94,32,306
25	9,51,468	1,74,62,535	4,20,000	48	42,89,253	1,27,53,282	22,21,85,588
26	9,46,711	1,80,70,232	4,20,000	29	44,42,203	1,32,08,029	23,53,93,617
27	9,41,977	1,86,99,076	4,20,000	17	46,00,473	1,36,78,602	24,90,72,219
28	9,37,267	1,93,49,803	4,20,000	10	47,64,250	1,41,65,553	26,32,37,772
29	9,32,581	2,00,23,177	4,20,000	6	49,33,726	1,46,69,451	27,79,07,223
30	9,27,918	2,07,19,983	4,20,000	4	51,09,099	1,51,90,884	29,30,98,107
Total	2,99,64,344	39,53,34,556	1,21,80,000	2,53,33,325	9,00,56,444	29,30,98,107	29,30,98,107

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

700 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.12 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹29.31 Cr
30-year generation	2,99,64,345 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 2,99,64,345 units, avoiding 24,570.8 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.66 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 700 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.